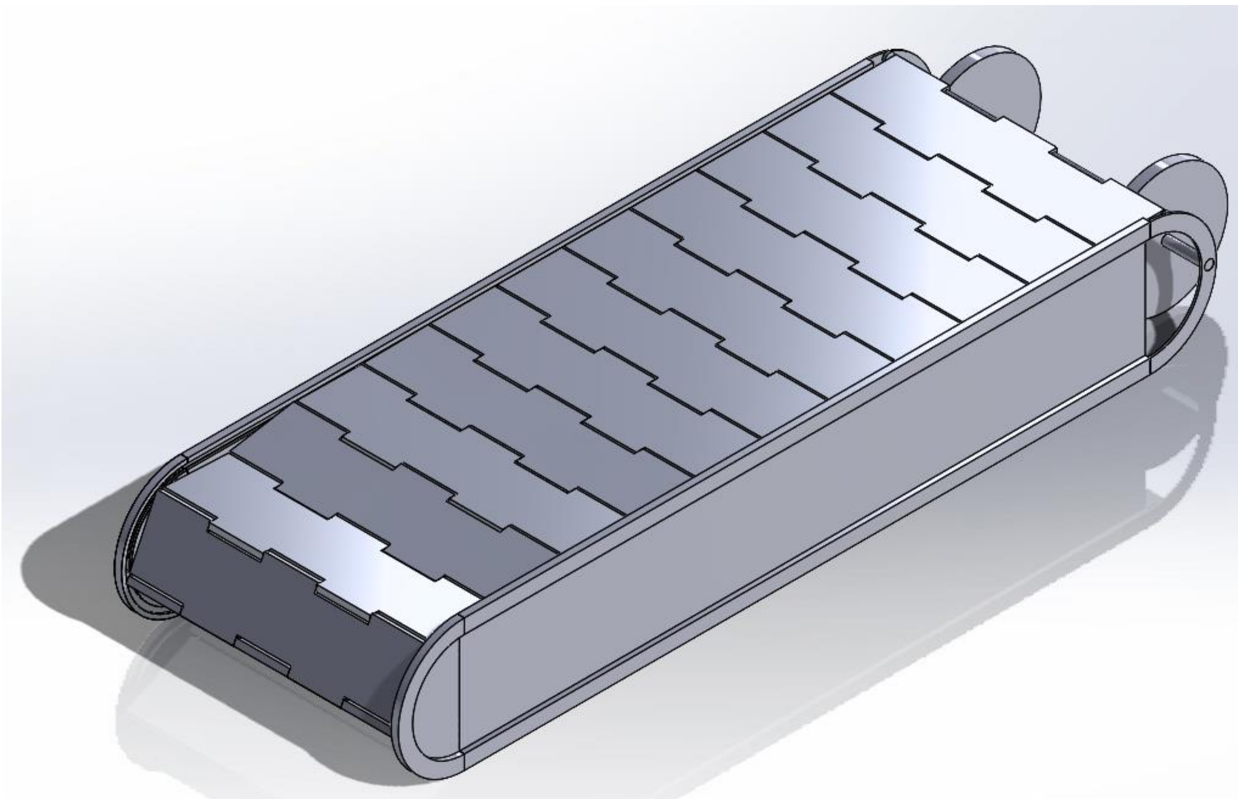
ARTEFACT 1

GIVEN DESIGN, OPERATION & FUNCTIONAL REQUIREMENTS



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UNIVERSTY OF TECHNOLOGY SYDNEY | FACULTY OF ENGINEERING AND INFORMATION TECHNOLOGY

41066 MECHANICAL SYSTEMS DESIGN STUDIO

GROUP 25

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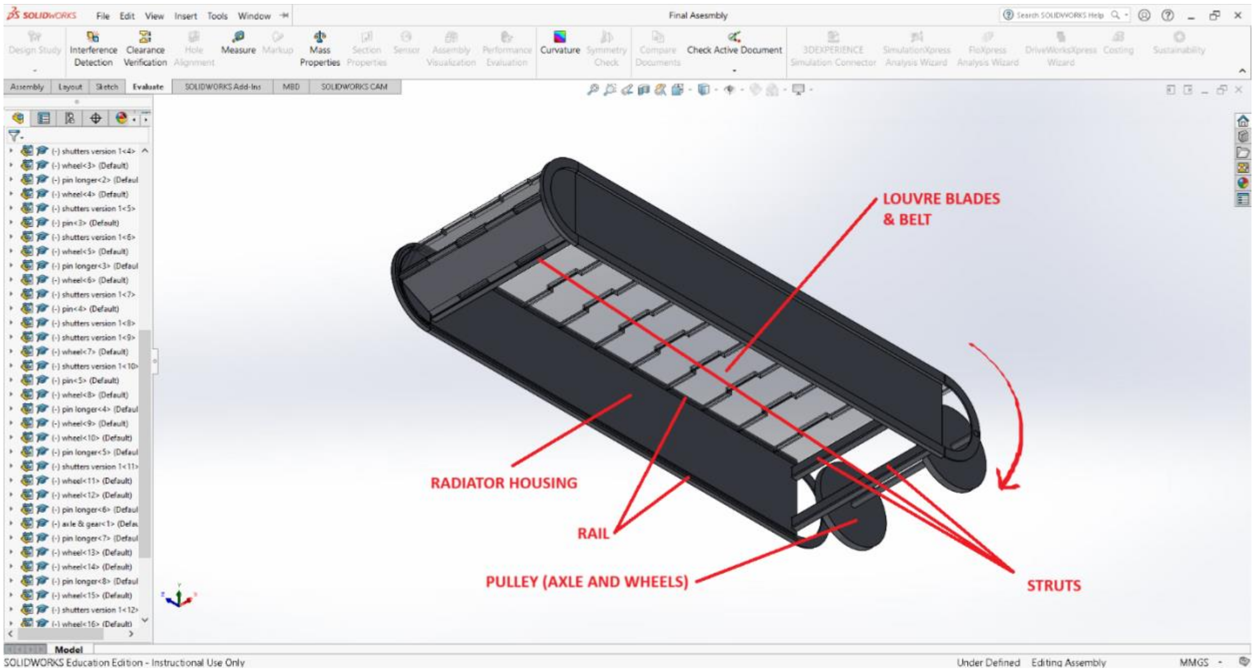
1 OPERATION (BEN)



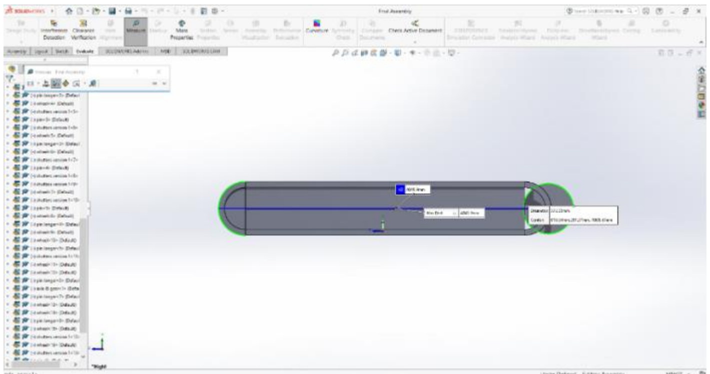
- Two side frames linked together via four struts to form the chassis of the louver system
- Radiator is mounted between the struts (Note: <u>within</u> louver system)
- Individual louver blades are linked together via pins upon which wheels are mounted to form louver belt
- Louver belt feeds into the rail channel in the louver chassis and articulate between active and inactive states along the rail
- Louver belt driven by pulley at one end that slots into the louver chassis

2 FUNCTIONAL REQUIREMENTS

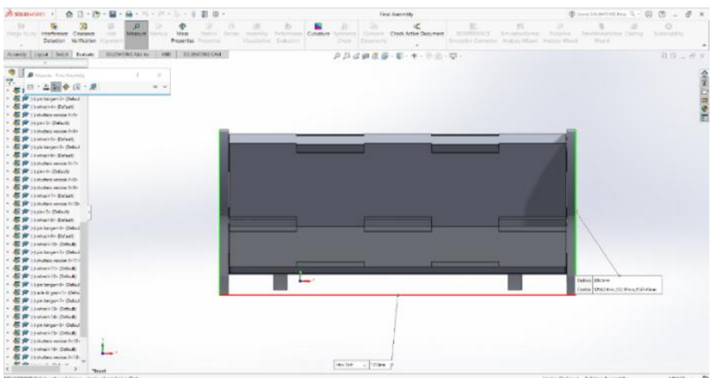
2.1 SIZE (BEN)

OUTER DIMENSIONS:

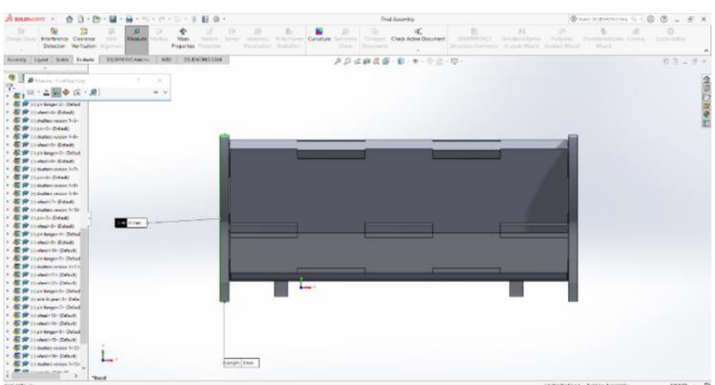
Length (4045.9mm/3080mm):



Width (1316/1250mm):

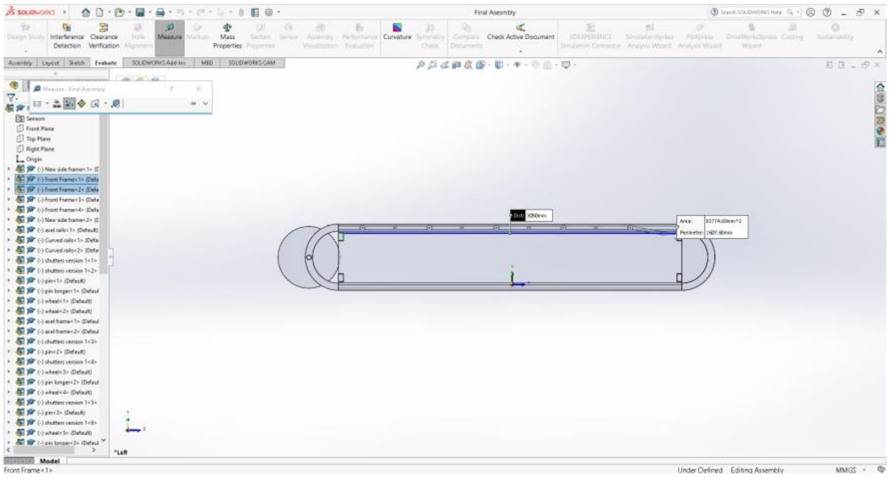


Height (613/460mm):

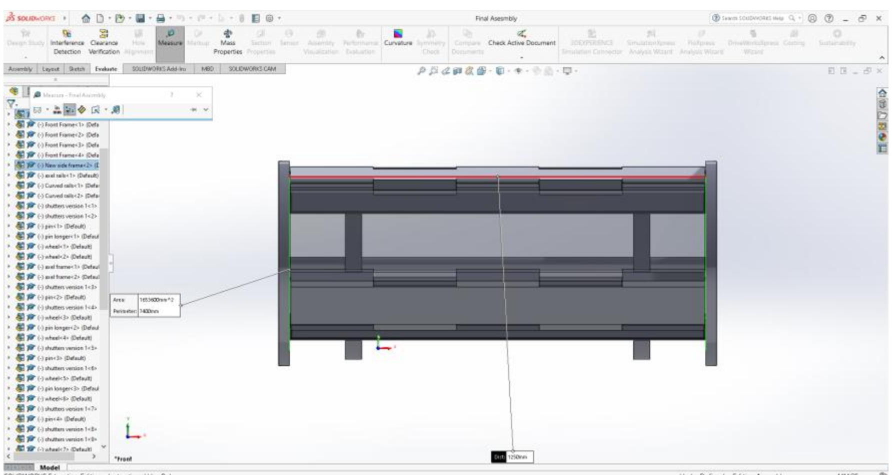


INNER DIMENSIONS (CONSTRAINED BY STRUTS)

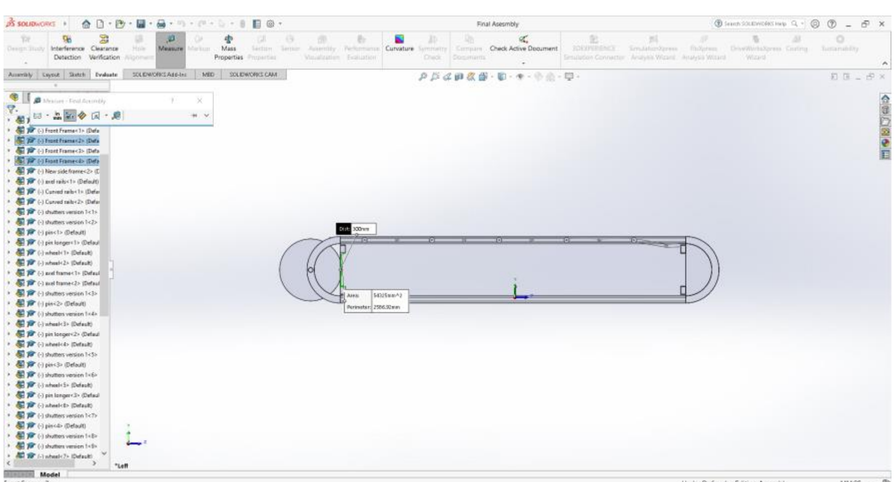
Length (3080mm):



Width (1250mm):



Height (300mm):



2.2 REQUIREMENT

(The louver system will need to cover a planar radiator surface of (W)450 mm x (L)600 mm and have a height < 70 mm.)

It is assumed that the radiator will be housed within the given louver system constrained by the 4 struts. The given louver system has outer dimensions (L)4069 x (W)1316 x (H)613mm and inner (constrained by the 4 struts) dimensions (L)3080 x (W)1250 x (H)300mm. The planar radiator surface can be accounted for within the louver design even after reducing the length by 80.5% (600.6mm) and the width by 64% (450mm), and the height must be reduced by AT LEAST 76.7% (69.9mm) to fit within the < 70mm height requirement. For the system to function at this reduction the dimensions of the louver, pulley and struts must be appropriately reduced. The louvers must undergo a dimension reduction for the blades to rotate about the curvature of the frame and fit on the rail within the frame in the first place. Similarly, the pulley; to align with the plane of the louver belt to minimize the length of the pulley belt while ensuring that the force on the louver belt when being pulled is along the rail to minimize friction and bending on the louver pins at the point where the wheels are attached. And the struts; to prevent obstruction to the radiator. The reduction in size will have a positive impact on weight and manufacturing costs but may adversely affect emissivity due to the reduction in surface area.

1.2 MASS (BEN)

(The louver system should have a mass density equivalent to/lower than that of louver systems used on past space missions.)

Table 9.1. Characteristics of Flight-Qualified Rectangular-Blade Louver Assemblies[a]

	OSC	Swales	Starsys
Blades	3 to 42		1 to 16
Open set points (°C)		0 to 40	–20 to 50
Open/close differential (°C)	10 or 18	10 or 18	14
Dimensions (cm)			
Length	20 to 110	27 to 80	8 to 43
Width	36 to 61	30 to 60	22 to 40
Height		6.4	6.4
Area (m^2)	0.07 to 0.6	0.08 to 0.5	0.02 to 0.2
Weight/area (kg/m^2)[b]	3.2 to 5.4	~ 4.5	5.2 to 11.6
Flight history	Nimbus, Landsat, OAO, ATS-6, Viking, GPS, SolarMax, AMPTE, SPARTAN, Hubble, Magellan, GRO, UARS, EUVE, TOPEX, GOES, MGS, MSP	XTE, Stardust	Rosetta[c], Quickbird[c] JPL:[d] Mariner, Viking, Voyager, Galileo, MLS, Magellan, TOPEX, NSCAT, Cassini, Seawinds

[a]This table contains representative values from past louver designs. Contact manufacturer for additional design possibilities or values for specific designs.
[b]Weight without sunshield.

Figure 1.1. Table 9.1 – Characteristics of Flight-Qualified Rectangular-Blade Louver Assemblies

Table 6.1. Alpha Radiator Characteristics

Characteristic	Description
Radiator	
Heat-rejection capacity	1250 W
Size	1.27 m × 3.18 m
Coating	Silver Teflon or quartz mirrors
Loop heat pipes	
Number of LHPs	4
Single pipe capacity at 65°C	>600 W @ 1 m adverse tilt
Evaporator length	457 mm
Condenser type	Direct condensation serial
Ground test elevation	>1 m with 1 failed LHP
Mechanisms	
Flex lines	Flex hose, 6.4 mm ID
Release device	G&H NEA
Hinge	Spherical bearing, torsion spring
Mass	
LHPs	9.5 kg
Radiator	10.2 kg w/silver Teflon, 11.3 kg w/quartz mirrors
Mechanisms	2.0 kg
Total	21.7 to 22.8 kg
Specific heat rejection	57.6 to 54.8 W/kg

Figure 1.2. Table 6.1 – Alpha Radiator Characteristics

In accordance with Gilmore (2002), the Starsys louver assembly with flight heritage possesses a weight/area of 5.2 to 11.6kg/m^2. This assembly was chosen and shall be the reference weight density due to sharing similarities with regards to louver count. The louver system to be designed should be less than this. The given louver design was analyzed and determined to possess a weight/density of approximately 8.87kg/m^2. Although this does fall within the reference range using the Starsys louver seembly, the sheer mass of the louver system is still 243kg. To place in perspective, an Alpha Radiator with flight heritage possessing a mass of 21.7 to 22.8kg, therefore the given louver system drastically exceeds it by more than ten times. However, as mentioned in the section about size, the given design is significantly larger and thicker than needs be for the radiator, so the reduction in size and the choice of a lighter material such as Al should see it fall soundly < 21.7kg in mass.

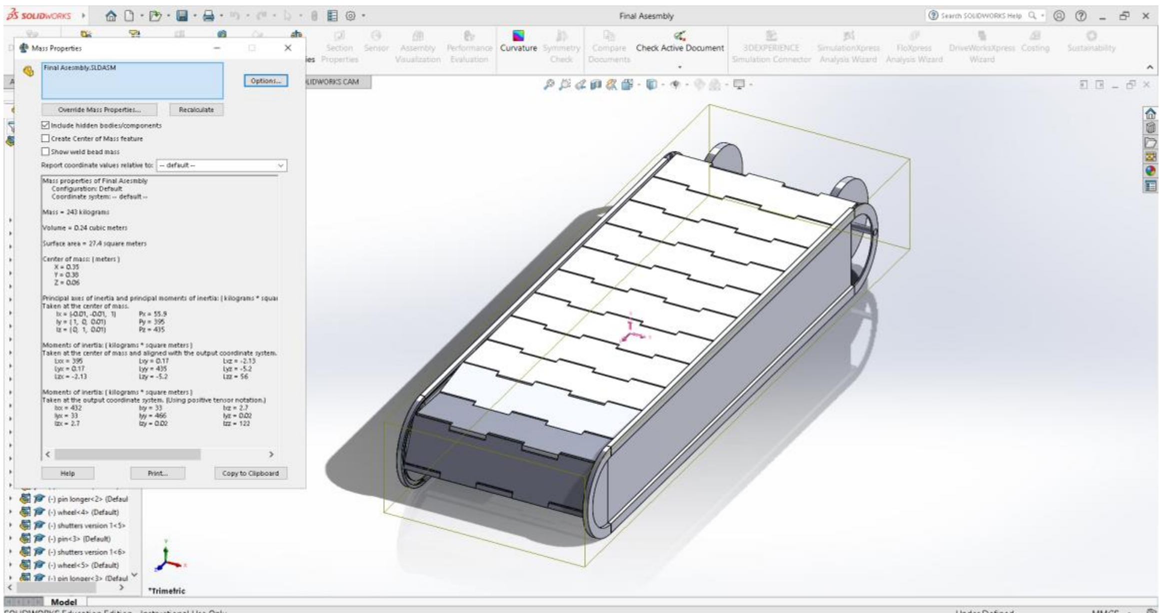


Figure 1.2. Mass properties of given louver system

1.3 THERMAL (ARTEFACT 2)

(The louver must vary heat rejection from the radiator and should reduce heat rejection by 10x from open-to-closed. The louver must be fully closed at T< 10°C and fully open at T>30°C)

1.4 POWER (NEO)

(Should operate passively without consuming electrical power.)

Through analysis of the given louver design, the actuation system seems to be powered by an electric DC or stepper motor. This component conflicts with the design requirements of being operated completely passively without consuming electrical power. This particular design constraint requires alternative actuation devices that can be powered by means other than electricity such as radiating heat energy from the sun or conducted heat energy from within the satellite system itself.

The power transmission from the motor to the shutters is not defined within the given CAD file so a solution must be chosen for the proper operation of the louver system. Some possible options are the wheels being pulley wheels that drive a belt throughout the louver system in order to slide the shutters along the slot or using a gear and chain system instead of the pulley belt system.

1.5 COST ANALYSIS (NHAN)

(Should be cost-effective to produce, with cost in-keeping with louver systems used in past space missions)

1.5.1 OVERVIEW

The cost analysis for the louver system focuses on evaluating the expenses associated with each part and material of the subsystems. The primary goal is to ensure the design is cost-effective while meeting the functional requirements for a short-term Low Earth Orbit (LEO) mission.

1.5.2 MATERIAL COSTS

1.5.2.1 Frame

- Material: aluminium
- Cost: approximately $3.80 per kg

The frame is made from aluminium due to its lightweight and strong properties. Aluminium is cost-effective and widely used in aerospace applications. The manufacturing process involves milling, which is relatively inexpensive for aluminium.

1.5.2.2 Struts

- Material: aluminium
- Cost: approximately $3.80 per kg

The struts are made from aluminium. The material choice ensures the structure is lightweight yet strong enough to support the louver system. The cost is kept low due to the ease of manufacturing aluminium struts.

1.5.2.3 Louver Blades

- Material: aluminium
- Cost:
  - Aluminium: approximately $3.80 per kg
  - High emissivity foil: $40 to $4000

The louver blades are made from aluminium to maintain a balance between weight and strength. The high emissivity coating on the blades adds to the cost but it enhances thermal control, which would be ideal for space applications such as low-Earth orbit satellites.

1.5.2.4 Pulley Axle

- Material: titanium
- Cost: $9.60 per kg

Titanium is chosen for the pulley axle due to its high strength and resistance to wear. Although titanium is more expensive than aluminium, its durability justifies the cost.

1.5.2.5 Pulley Wheels

- Material: nylon
- Cost: approximately $4 per kg and $8 per kg for nylon 6 and nylon 66, respectively

Nylon is used for the pulley wheels due to its lightweight and low friction properties. The cost of nylon is relatively low, making it a cost-effective choice for the wheels.

1.5.2.6 Louver Pins

- Material: aluminium
- Cost: approximately $3.80 per kg

The louver pins are made from aluminium to keep the weight low and ensure ease of assembly.

1.5.2.7 Louver Wheels

- Material: nylon
- Cost: approximately $4 per kg and $8 per kg for nylon 6 and nylon 66, respectively

The louver wheels are made from nylon to ensure low friction and lightweight properties, keeping the cost low.

1.5.2.8 Louver Belt Chord

- Material: nylon
- Cost: approximately $4 per kg and $8 per kg for nylon 6 and nylon 66, respectively

The louver belt chord is made from nylon to provide flexibility and strength. The cost is kept low due to the availability of the material and ease of manufacturing.

1.5.2.9 Motor Hub

- Material: aluminium
- Cost: approximately $3.80 per kg

Aluminium is lightweight and cost-effective, suitable for the motor hub.

1.5.2.10 M4 Nuts, Bolts, and Washers

- Material: titanium
- Cost:
    - M4 nuts: $23.75 per 10 nuts → $68.88 for 29 nuts
    - M4 bolts: $38.09 per 10 bolts → $76.18 for 20 bolts
    - M4 washers: $16.31 per 10 washers → $26.10 for 16 washers

Fasteners are critical in ensuring structural integrity and reliability in the louver system. Titanium offers exceptional strength-to-weight ratio, corrosion resistance, thermal stability, and durability, which are essential for withstanding the harsh conditions of space.

1.5.3 COST EFFECTIVENESS

- Aluminium: widely used in aerospace for its balance of strength, weight, and cost. It is less expensive than titanium and provides sufficient durability for short-term missions.
- Carbon composite: offers superior strength and thermal properties, essential for the louver blades. The high emissivity coating ensures efficient thermal management.
- Nylon: chosen for its low cost and adequate performance in low-stress components like wheels.
- Titanium: used selectively to minimise costs while ensuring critical components like the axle are durable and wear-resistant.

The selected materials and design choices aim to balance cost and performance, ensuring the louver system is both effective and economical for a short-term LEO mission. By using cost-effective materials like aluminium and nylon and reserving more expensive materials like titanium for critical components, the overall cost is minimised while maintaining functionality and reliability.

1.5.4 POSSIBLE OVERSIGHTS/COST BLOWOUTS

- High emissivity coatings: applying high emissivity coatings often involves precise surface preparation, multiple coating layers, and controlled environments to ensure proper adhesion and performance. This process can be time-consuming and expensive.
- Material waste: the high cost of raw materials and potential waste during manufacturing can contribute to cost overruns.
- Quality control: ensuring the uniformity and integrity of these materials and coatings requires rigorous quality control measures which could impact to the overall cost.
- Assembly and integration: misalignment or improper assembly can cause functional issues, which could potentially lead to additional costs for rework or replacement.
- Environmental factors: exposure to micrometeoroids and thermal cycling in space can damage components.

1.6 COMPATIBILITY (BEN)

(Must meet applicable space standards)

Designing a louver for operation in space on a satellite intended for a short-term Low Earth Orbit (LEO) mission involves several standards to ensure functionality, reliability, and performance. A summary of such standards are as follows:

- NASA-STD-5001: Defines structural design and test factors of safety, in our case with regards to satellite components, specifically the louvers
- ECSS-E-ST-31-01C (European Cooperation for Space Standardization): Outlines thermal design and analysis for space systems, including passive thermal control components which regards louvers.
- ISO 14644-1: Focused on cleanliness, it is essential that louvers meet contamination control requirements, as particulates could affect thermal performance.
- ASTM E595: Materials used for louvers must have low outgassing properties to prevent contamination of sensitive spacecraft components.

Essentially, the louver must be designed for survival under extreme circumstances; one such being temperature variations, typically from -100°C to +120°C in LEO and capable of radiating this extreme heat and shielding this extreme cold. Micrometeoroid impact is another such consideration that the louver must be designed to withstand or otherwise have redundancy from.

1.7 RELIABILITY (NEO)

(Components will need to be durable and withstand the extreme conditions of space.)

Within the harsh environment of the vacuum of Space, the louver must endure extreme conditions throughout each run which can last between 10 to 30 days whilst experiencing temperatures ranging from $-65℃\ to\ 125℃$. (Jeannette Plante, 2004) It is also important to account for the Factor of Safety to ensure that each component of the louver will perform at acceptable levels throughout the mission timespan.

The material choices will greatly influence the performance and reliability of the system, which means materials with higher tolerance against deformation due to heat will be preferable, whilst keeping the other subsystem requirements fulfilled. Some viable options for high performance materials include space grade steel, aluminium and titanium (Finckenor1), as well as some polymers like Polybenzimidazole (PBI previously used in manned space mission) and Polyetherimide (PEI). These options will need to be evaluated against the other design function requirements such as cost and mass to determine the optimal fit.

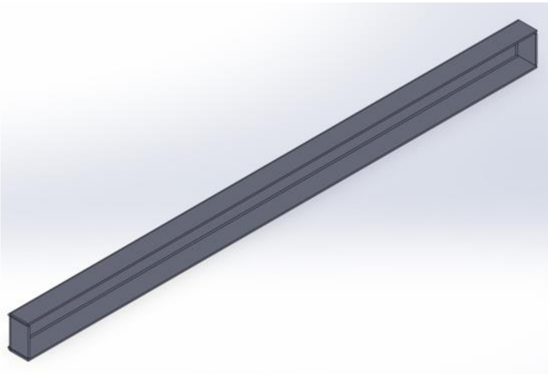
The actuation mechanism should be operable at any given circumstance that the satellite will encounter so backup measures such as backup electrically powered systems may be options that will need to be considered.

1.8 MANUFACTURE (NEO - ((DIS)ASSEMBLY/DARCY - (BoM))

1.8.1 MANUFACTURE

1.8.1.1 Frame

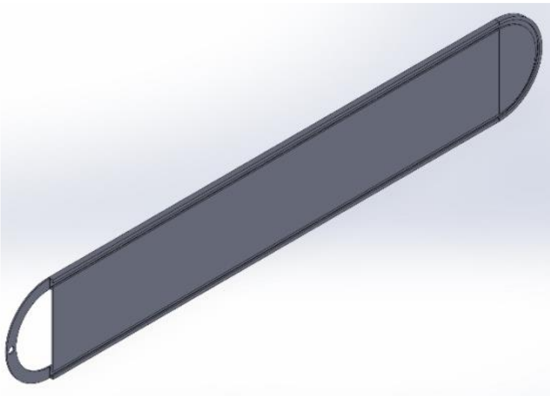
Struts



Strut will be relatively easy to manufacture, whether it sheet bending or machined, depending on material choice and cost benefit analysis.

Material should be strong metal as bend resistance is important. Aluminium or titanium may serve this purpose.

Side Frame



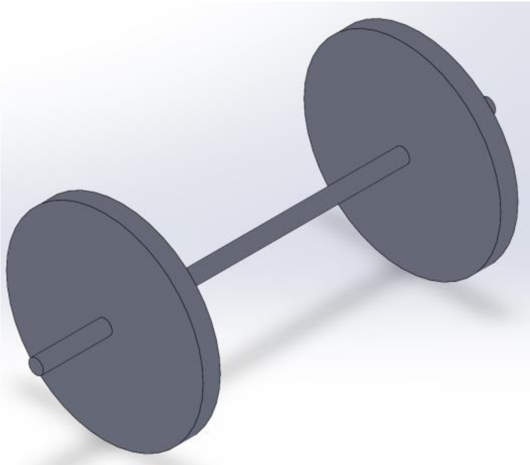
The frame presents manufacturing difficulties as the required cutout for the wheels to run in would be difficult to machine or cast. A future design would have the frame made from multiple pieces that would be fastened together.

Material would be a lightweight, strong metal, such as aluminium or titanium. Part is relatively large, so would need to keep cost in mind too. It is also important to remember that fasteners would be tapped into this piece, so softness considerations are also important to prevent galling.

1.8.1.2 Actuator

NA

1.8.1.3 Pulley

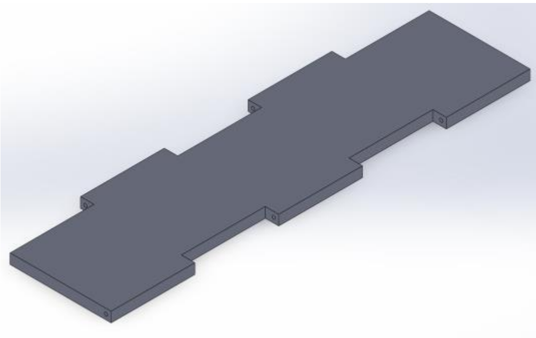


The axle should be strong metal to prevent wearing, though interaction between friction surfaces is important to consider to prevent damage. Titanium is such a material.

Wheels can be lightweight and rigid, such as a temperature resistant plastic. Nylon is a temperature resistant, lightweight plastic that has a low coefficient of friction to reduce energy losses.

1.8.1.4 Louver Belt

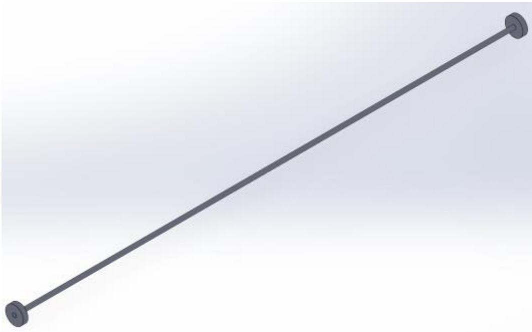
Louver Blade



Louver blade material should be lightweight as it composes high percentage of overall volume of object. It has some force subject to it, but is mostly used to block radiation from passing, so should be highly emissive.

Carbon composite or aluminium may be applicable, with an emissive coating such as a PVD coating or high emissivity paint or foil.

1.8.1.5 Louver Wheel & Axle



Axle should be a strong metal, to prevent bending, similar to pulley axle.

Wheels should be a low friction material to assist in sliding through the frame, to minimize energy required of motor assembly. Nylon satisfies these needs.

1.8.2 BILL OF MATERIALS

Preliminary Bill of Materials, with assumed material choice and short description. Subsystems are identified by colours:

Subsystem	*Colour*	*Description*
Actuator	Purple	The effective motor of the system
Frame	Orange	Sides and struts
Louver Belt	Blue	Louver Blades, Louver axles and rollers
Pulley System	Green	Main pulleys, cable and pulley axle

Name	Part No#	Description	Material	Quantity
Side Frame	1	Solid Milled piece	Aluminium	2
Struts	2	Connects Side frame, for rigidity	Aluminium	4
Louver Blades	3		Aluminium	14
Pulley Axle	4		Titanium	1
Pulley	5		Nylon	2
Louver Pin	6	Threaded end, capped end	Aluminium	14
Louver Wheel	7		Nylon	28
Louver Belt chord	8	Runs from last louver blade to first, around pulley	HR125 Rubber	2
Strut Bolt & Nut	9	M4 nut, bolt and washer set, 4 per strut	Titanium	16
Louver Pin Nut	10	M4 nut	Titanium	9
Motor Hub	11	Frame for motor, attaches to chassis	Aluminium	1
Chosen Motor	12			1
Motor Fasteners	13	M4 nut and bolt	Titanium	4

1.8.3 ASSEMBLY

(Should be easy to assemble by a single person using only basic tools.)

Identify parts/fasteners that can be combined or eliminated: Many of the pins and wheels can be combined together for easier manufacturing and assembly. (the shutter pins and wheels), the pins can also all be the same length ensuring more similar parts as well as more uniformity during shutter actuation.

Make sure to include any inefficiencies in this process that we can cut for later artefacts

Assembly and Disassembly

1. Modular design:
    a. The actuating mechanism (axle and wheel) can be disassembled from the louver chassis
    b. The louver blades can be removed from the chassis
2. Maximize symmetry and near symmetry (poka-yoke):
    a. Fully assembled has near-symmetry along two axes and is symmetrical along one axis
    b. Disassembling the actuating mechanism from the louver chassis results in symmetry along all axis for both the louver chassis and actuating mechanism
3. Clear visible features of alignment and positioning
    a. Rail for louver blades
    b. Slot to mount actuating mechanism
4. Minimize number of operations, directions and orientation of parts
    a. Bi-directional displacement of louver array
    b. Louver chassis consists of two extrusions connected by four struts
    c. Deployed orientated laying down

ASSEMBLY								
Part #	Part Name	Qty	Handling Code	Handling Time/Qty (s)	Insertion Code	Insertion Time/Qty (s)	Total Times (s)	Comments
1	Side Frame	2	N/A	2	N/A	0	4	
2	Struts (including nut and bolts)	4	N/A	2	N/A	8	40	
3	Pulley (including belt)	2	N/A	5	N/A	20	50	
4	Pulley Axle	1	N/A	2	N/A	10	12	
5	Shutter Wheel	28	N/A	1	N/A	3	112	
6	Shutter Pin	14	N/A	2	N/A	5	98	
7	Shutter	14	N/A	1	N/A	0	14	
8	Shutter Assembly	1	N/A	1	N/A	10	11	
9	Motor Assembly	1	N/A	10	N/A	45	55	
				26.0		**101.0**	**396.0**	
							6.6	mins

1.8.3.1 PROCEDURE

Assumptions:

- The two side frames come manufactured as represented

Subsystem: Louver Frame

1. Position two side frames in parallel and mirroring
2. Fix the side frames together at the four fixture points with four struts

Subsystem: Louver Belt

3. Link all louver blades to each other
4. Fix each louver belt to each other with pin
5. Attach louver wheels to each pin and secure in place forming the louver belt
6. Slot louver belt into rail channeled in frame

Subsystem: Pulley

7. Align pulley wheels within frame concentrically to slot
8. Slot in pulley axle and secure
9. Equip belt to pulley and link to louver belt

Subsystem: Motor

10. Align motor hub to chassis and attach
11. Align motor to motor hub and attach, ensuring motor axle mates with the pulley wheel

1.8.4 DIASSEMBLY

The disassembly will be considered from to perspectives:

- Into Subsystem
- Into Individual Parts

1.8.4.1 SUBSYSTEM PROCEDURE

It is useful to identify procedures for removing subsystems, as it may not be individual parts being replaced, but the subsystem as a whole.

Motor Subsystem

1. Remove fasteners fixing motor to motor hub, remove motor, ensuring disconnect from pulley axle
2. Remove Motor hub fasteners and then motor hub from chassis

Louver Subsystem

1. Remove belt from pulleys

2. Unfasten chord from last louver blade
3. Slide entire Louver belt out of frame

Pulley Subsystem

1. Remove pulley axle from frame and wheels.
2. Pulley wheels are now free

Frame Subsystem

1. Frame is now the only remaining subsystem

DISASSEMBLY - Subsystems								
Part #	Part Name	Qty	Handling Code	Handling Time/Qty (s)	Insertion Code	Removal Time/Qty (s)	Total Times (s)	Comments
1	Motor Assembly	1	N/A	20	N/A	40	60	
2	Louvre Belt	1	N/A	10	N/A	20	30	
3	Pulley	1	N/A	5	N/A	30	35	
4	Frame	2	N/A	0	N/A	0	0	
				35.0		**90.0**	**125.0**	
							02:05	mins

1.8.4.2 INDIVIDUAL COMPONENTS PROCEDURE

If an individual part needs replacement the following procedures may be taken to strip the whole assembly to individual parts.

Motor Subsystem

1. Remove fasteners fixing motor to motor hub, remove motor, ensuring disconnect from pulley axle
2. Remove Motor hub fasteners and then motor hub from chassis

Louver Subsystem

1. Remove belt from pulleys
2. Unfasten chord from last louver blade
3. Slide entire Louver belt out of frame.
4. Unfasten chord from first louver blade
5. Remove fasteners from each louver axle
6. Remove each louver wheel and slide louver axles out
7. Louver blades are now free

Pulley Subsystem

3. Remove pulley axle from frame and wheels.

4. Pulley wheels are now free

Frame Subsystem

8. Remove fasteners from struts
9. Struts and side frames are now free

DISASSEMBLY - Individual Components								
Part #	Part Name	Qty	Handling Code	Handling Time/Qty (s)	Insertion Code	Removal Time/Qty (s)	Total Times (s)	Comments
1	Side Frame	2	N/A	2	N/A	0	4	
2	Struts (including nut and bolts)	4	N/A	2	N/A	8	40	
5	Pulley (including belt)	2	N/A	5	N/A	5	20	
4	Pulley Axle	1	N/A	2	N/A	10	12	
3	Louvre Blade	14	N/A	1	N/A	0	14	
6	Louvre Pin	14	N/A	2	N/A	2	56	
7	Louvre Wheel	28	N/A	1	N/A	1	56	
8	Motor Assembly	1	N/A	5	N/A	30	35	
				20.0		**56.0**	**237.0**	
							03:57	mins

1.9 MAINTENANCE (NEO)

(Should be repairable in space by a human or robot, with the ability to disassemble and reassemble.)

The design of the louver is quite simplistic which will allow for ease of maintenance as subsystems can be made to be modular, therefore, easily to replace. Also, each of the components can be made to stronger and more durable thanks to the simplistic nature of the design, meaning there are lower amounts and less intricate individual parts.

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